

Task 1 Report: Data Cleaning and Visualization

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Domain: Data Science Internship

Project: Flight Delay Data Analysis

Title:

Flight Delay Data Analysis

Introduction:

In this project, I worked on a flight delay dataset to understand how delays happen and what factors affect them. The main goal was to clean the data and create visualizations to get useful insights.

Dataset Description:

The dataset contains details about different flights such as airline name, source and destination, scheduled and actual times, delay minutes, delay reasons, and distance.

Data Cleaning:

Before analysis, I cleaned the dataset by:

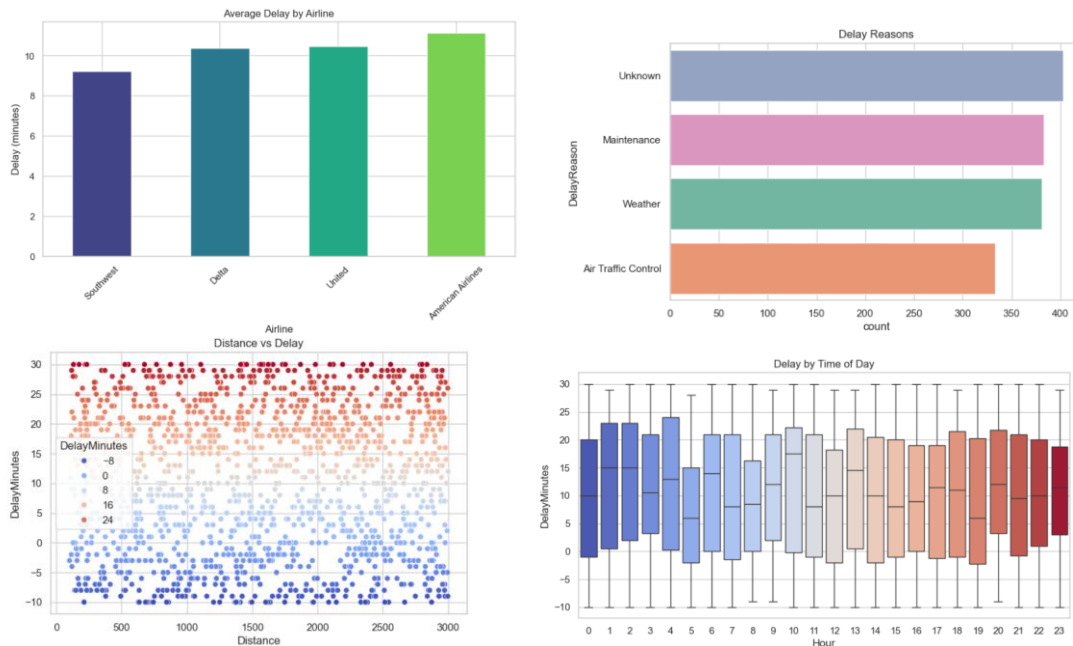
- Checking for missing values
- Filling missing values in **DelayReason** with "Unknown"
- Removing duplicate rows
- Converting date columns into proper datetime format
- Creating a new column **Hour** from departure time

Data Visualization:

I created different graphs to understand the data better:

- Bar chart to compare delay across airlines
- Histogram to see how delay minutes are distributed
- Count plot to check common delay reasons
- Scatter plot to analyze distance vs delay

- Count plot for cancelled flights
- Box plot to see delay based on time of day



Insights:

- Some airlines have more delays than others
- Weather and air traffic are common delay reasons
- Longer distance flights may have slightly higher delays
- Some flights have negative delay values, meaning they arrived early
- Delays vary depending on the time of the flight

Conclusion:

This project helped me understand how data cleaning and visualization work in real datasets. It also showed how useful insights can be obtained from simple analysis.

Tools Used:

- Python
- Pandas
- Matplotlib
- Seaborn